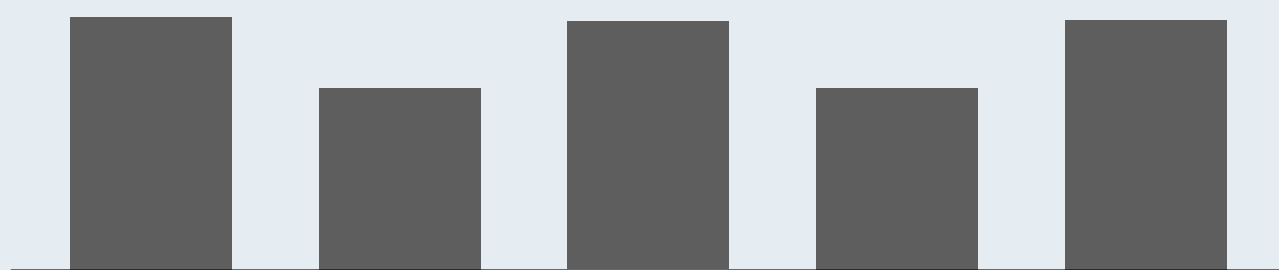
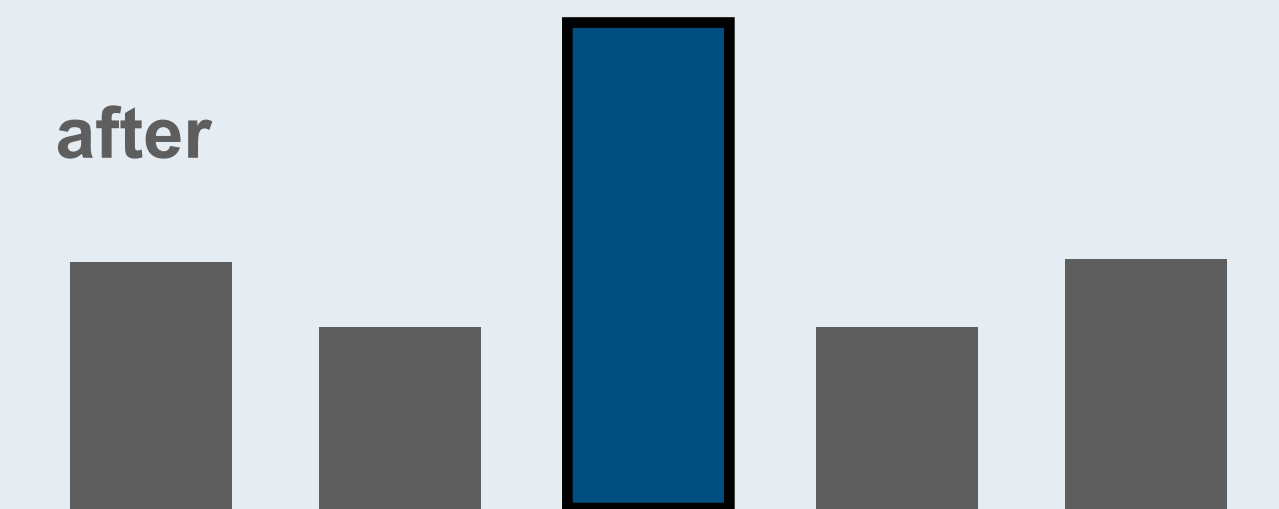


Bayesian update contraction

before



$P(h) \leftarrow P(h | d)$ observe data d



mass concentrates on
data-supporting hypothesis

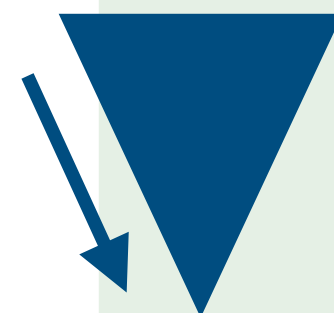
Bayesian Inverse-Bayesian (BIB)

contraction

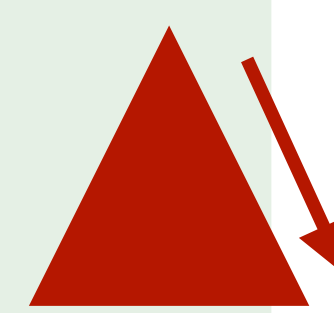


Criticality

relaxation



realize



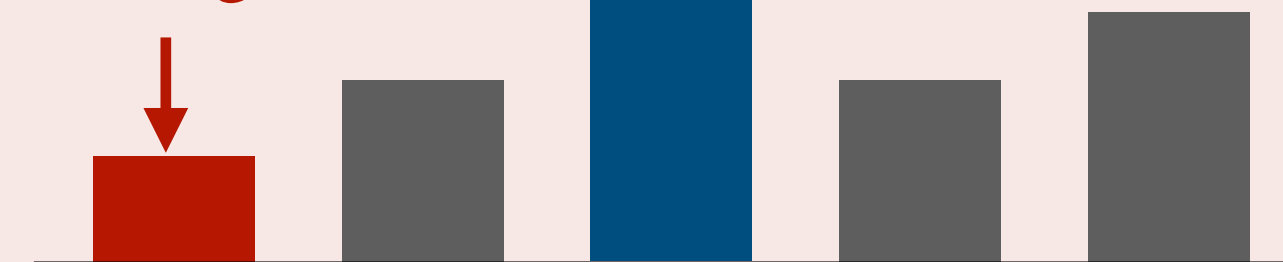
$$P(T_{\text{argmax}}) \sim T^{-\alpha}$$
$$\alpha \approx 1.43 \text{ (universal)}$$

Universal critical dynamics

Inverse Bayesian relaxation

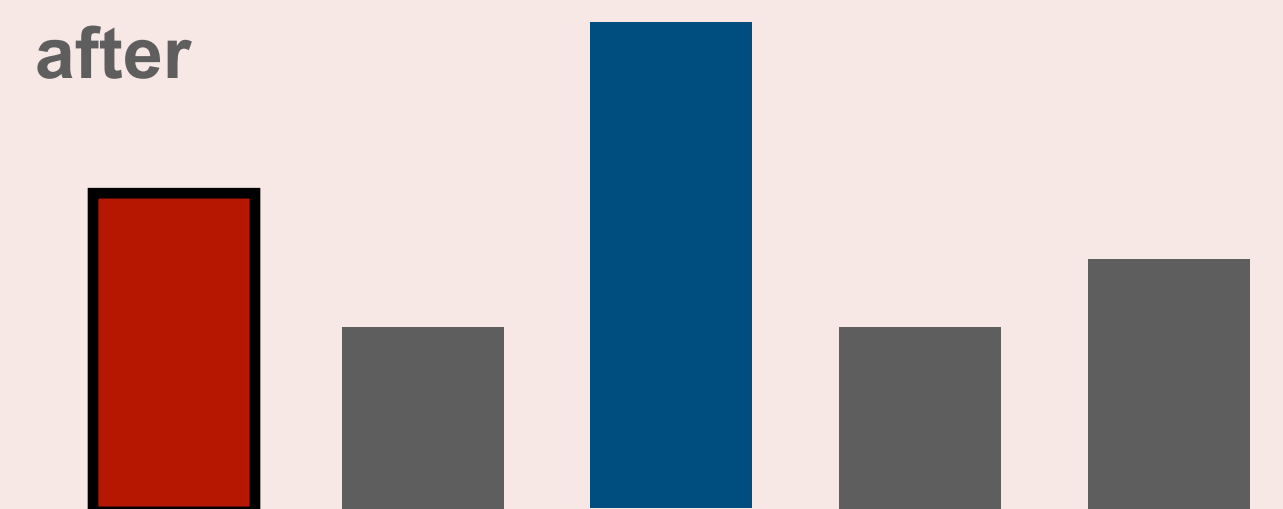
before

$$i^* = \arg \min P(h)$$



$P(d | h) \leftarrow P(d)$ replace by
empirical p_t

after



new hypothesis injected
reflects recent data flow

cycle continues: new data $\rightarrow$ Bayesian update $\rightarrow \dots$